

Project Design Phase

Proposed Solution Template

Date	15 February 2025	Maximum Marks	2 Marks
	—		Credit Card Approval Prediction

Project team shall fill the following information in the proposed solution template.

S.No. 1	Parameter: Problem Statement (Problem to be solved)
Description:	Financial institutions face significant challenges in accurately evaluating credit card applications. Manual review processes are time-consuming, inconsistent, and prone to human bias. Approving high-risk applicants leads to increased default rates and financial losses, while rejecting eligible applicants results in lost revenue and poor customer experience. There is a pressing need for an automated, data-driven system that can reliably predict whether a credit card application should be approved or rejected — based on applicant demographics, financial history, and behavioral attributes — thereby reducing risk and improving operational efficiency.
S.No. 2	Parameter: Idea / Solution Description
Description:	The proposed solution is a Machine Learning-based Credit Card Approval Prediction System. Historical applicant data (including age, income, employment status, credit history, debt levels, and prior defaults) is used to train supervised classification models such as Logistic Regression, Random Forest, and XGBoost. The trained model is integrated into a user-friendly web application (built with Flask/Streamlit) where bank officers or applicants can input details and receive an instant approval/rejection prediction along with a confidence score. The pipeline includes data preprocessing, feature engineering, model training, evaluation (using accuracy, precision, recall, F1-score, and AUC-ROC), and deployment.
S.No. 3	Parameter: Novelty / Uniqueness
Description:	Unlike rule-based systems that rely on fixed thresholds, this solution learns complex, non-linear patterns from data. Key novelties include: (a) Ensemble model comparison — multiple algorithms are benchmarked to select the best performer; (b) Explainability — SHAP (SHapley Additive exPlanations) values are used to make predictions interpretable for compliance and auditing; (c) Bias mitigation — fairness-aware preprocessing ensures the model does not discriminate based on gender or ethnicity; (d) Real-time inference — the deployed model provides instant predictions via a REST API, making it production-ready for integration with existing banking systems.
S.No. 4	Parameter: Social Impact / Customer Satisfaction

Description:	The solution creates measurable social and economic value: (a) Financial Inclusion — deserving applicants from underserved segments who are otherwise rejected due to manual bias receive fair evaluation; (b) Reduced Default Risk — banks experience fewer non-performing assets, stabilising the broader financial system; (c) Faster Processing — applicants receive near-instant decisions instead of waiting days/weeks, improving satisfaction; (d) Transparency — explainable AI helps customers understand why they were approved or rejected, increasing trust; (e) Cost Savings — automation reduces manual workload for bank staff, cutting operational costs.
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S.No. 5 Parameter: Business Model (Revenue Model)

Description:	The system can be monetised through multiple channels: (a) SaaS Subscription — banks and NBFCs pay a monthly/annual subscription fee to access the prediction platform via API or web dashboard; (b) Pay-per-Prediction — a micro-transaction model where institutions are charged per API call, suitable for smaller lenders; (c) White-label Licensing — the solution is packaged and licensed to fintech companies under their branding; (d) Consulting & Integration Services — one-time setup, customisation, and integration fees; (e) Data Analytics Add-on — premium dashboards offering portfolio-level risk analytics, trend reports, and regulatory compliance summaries sold as an upsell.
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S.No. 6 Parameter: Scalability of the Solution

Description:	The architecture is designed for horizontal scalability: (a) Cloud Deployment — the model is containerised using Docker and orchestrated with Kubernetes on AWS/GCP/Azure, enabling auto-scaling based on demand; (b) Batch & Real-time Modes — supports both bulk processing of thousands of applications overnight and real-time single predictions via REST API; (c) Model Retraining Pipeline — an automated MLOps pipeline (using MLflow/Airflow) retrains the model periodically on new data, ensuring accuracy over time; (d) Multi-region Support — the platform can be configured for different countries by adapting feature sets and regulatory requirements; (e) Microservices Architecture — individual components (data ingestion, preprocessing, inference, reporting) are independently deployable, allowing teams to scale or update each module separately.
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